

- 26.** If q is the mean proportional between p and r , then $\frac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}}$ is equal to
 (a) p^2q^3 (b) q^3 (c) q^4 (d) $p^2r^2q^4$
- 27.** The monthly incomes of A and B are in the ratio 4 : 3. Each of them saves ₹ 600. If the ratio of the expenditure is 3 : 2, then what is the monthly income of A ?
 (a) ₹ 2400 (b) ₹ 1800 (c) ₹ 2000 (d) ₹ 3600
- 28.** Three numbers are in the ratio 3 : 2 : 5 and the sum of their squares is 1862. What are the three numbers?
 (a) 18, 12, 30 (b) 24, 16, 40
 (c) 15, 10, 25 (d) 21, 14, 35
- 29.** The speeds of three cars are in the ratio 4 : 3 : 2. What is the ratio between the times taken by the cars to cover the same distance?
 (a) 2 : 3 : 4 (b) 3 : 4 : 6
 (c) 1 : 2 : 3 (d) 4 : 3 : 2
- 30.** The ratio between the ages of A and B is 2 : 5. After 8 yr, their ages will be in the ratio 1 : 2. What is the difference between their present ages?
 (a) 20 yr (b) 22 yr (c) 24 yr (d) 25 yr
- 31.** Let y is equal to the sum of two quantities of which one varies directly as x and the other inversely as x . If $y = 6$ when $x = 4$ and $y = 10/3$, when $x = 3$, then what is the relation between x and y ?
 (a) $y = x + (4/x)$ (b) $y = -2x + (4/x)$
 (c) $y = 2x + (8/x)$ (d) $y = 2x - (8/x)$
- 32.** If ₹ 8400 is divided among A, B and C in the ratio $\frac{1}{5} : \frac{1}{6} : \frac{1}{10}$, what is the share of A ?
 (a) ₹ 1800 (b) ₹ 3000 (c) ₹ 3600 (d) ₹ 4000
- 33.** A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it was divided in the ratio of 2 : 3 and there by Q loses ₹ 10. What was the amount?
 (a) ₹ 250 (b) ₹ 300 (c) ₹ 350 (d) ₹ 400
- 34.** 20 L of a mixture contains milk and water in the ratio 4 : 3. If 6 L of this mixture are replaced by 6 L of milk, the ratio of milk to water in the new mixture will become
 (a) 7 : 3 (b) 8 : 3 (c) 9 : 7 (d) 4 : 6
- 35.** Two vessels are full with milk and water in the ratios 1 : 3 and 3 : 5, respectively. If both are mixed in the ratio 3 : 2, what is the ratio of milk and water in the new mixture?
 (a) 4 : 15 (b) 3 : 7
 (c) 6 : 7 (d) None of these
- 36.** A mixture contains milk and water in the ratio 5 : 1. On adding 5 L of water, the ratio of milk and water becomes 5 : 2. What is the quantity of milk in the original mixture?
 (a) 5 L (b) 25 L
 (c) 27.5 L (d) 32.5 L
- 37.** If x varies as the m th power of y , y varies as the n th power of z and x varies as the p th power of z , then which one of the following is correct?
 (a) $p = m + n$ (b) $p = m - n$
 (c) $p = mn$ (d) None of these
- 38.** The wages of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : 2. What was the original wage bill of the factory, if the present bill is ₹ 5000?
 (a) ₹ 4000 (b) ₹ 6000
 (c) ₹ 8000 (d) None of these
- 39.** The ratio of A to B is $x : 8$ and the ratio of B to C is $12 : z$. If the ratio of A to C is $2 : 1$, then what is the ratio of $x : z$?
 (a) 2 : 3 (b) 3 : 2 (c) 4 : 3 (d) 3 : 4
- 40.** A bag contains ₹ 112 in the form of ₹ 1, 50 paise and 10 paise coins in the ratio 3 : 8 : 10. What is the number of 50 paise coins?
 (a) 112 (b) 108 (c) 96 (d) 84
- 41.** In a class, the number of boys is more than the number of girls by 12% of the total students. What is the ratio of number of boys to that of girls?
 (a) 11 : 14 (b) 14 : 11
 (c) 28 : 25 (d) 25 : 28
- 42.** A sum of ₹ 53 is divided among A, B, C in such a way that A gets ₹ 7 more than what B gets and B gets ₹ 8 more than what C gets. The ratio of their shares is
 (a) 25 : 18 : 10 (b) 6 : 7 : 8
 (c) 12 : 14 : 9 (d) 15 : 8 : 30
- 43.** A person P started a business with a capital of ₹ 2525 and another person Q joined P after some months with a capital of ₹ 1200. Out of the total annual profit of ₹ 1644, P 's share was ₹ 1212. When did Q join as partner?
 (a) After 2 months (b) After 3 months
 (c) After 4 months (d) After 5 months
- 44.** Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75%, respectively. What will be the ratio of increased seats?
 (a) 2 : 3 : 4 (b) 6 : 7 : 8
 (c) 6 : 8 : 9 (d) 5 : 7 : 12

- 45.** The ratio between the number of passengers travelling by I and II class between the two railway stations is 1:50, whereas the ratio of I and II class fares between the same station is 3:1. If on a particular day, ₹ 1325 were collected from the passengers travelling between these stations, then what was the amount collected from the II class passengers?

(a) ₹ 750 (b) ₹ 1000
(c) ₹ 850 (d) ₹ 1250

- 46.** If x varies inversely as the square of y in such a way that, if $x = 1$, then $y = 6$.

I. If $y = 3$, then $x = 4$ II. If $y = 6$, then $x = 1$

Which of the following options is correct?

(a) Only I (b) Both I and II
(c) Only I (d) Neither I nor II

- 47.** If $a : b = c : d = e : f = 1 : 2$, then

I. $\frac{3a+5c+7e}{3b+5d+7f} = \frac{1}{3}$ II. $\frac{a^2+c^2+e^2}{b^2+d^2+f^2} = \frac{1}{2}$

Which of the following option is correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 48.** A cat takes 5 leaps for every 4 leaps of a dog but 3 leaps of the dog are equal to 4 leaps of the cat. Now, the

I. Ratio of the speeds of the cat to that of the dog is 15 : 16.

II. Ratio of the distance of the cat to that of the dog is 15:16, covered in 30 min.

Which of the following option is correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 49.** Determine the ratio of the number of people having characteristic X to the number of people having characteristic Y in a population of 100 subjects from the following table:

Having X and Y	10
Having X but not Y	30
Having Y but not X	20
Having neither X nor Y	40

(a) 4:3 (b) 3:2 (c) 1:2 (d) 2:3

- 50.** Fresh grapes contain 90 percent water by weight while dried grapes contain 20 percent water by weight. What is the weight of dry grapes contain 20 percent water by weight. What is the weight of dry grapes available from 20 kg of fresh grapes?

(a) 2 kg (b) 2.4 kg
(c) 2.5 kg (d) None of these

PREVIOUS YEARS' QUESTIONS

- 51.** If $x : y = 7 : 5$, then what is the value of $(5x - 2y) : (3x + 2y)$? ☑ 2012 I

(a) $5/4$ (b) $6/5$ (c) $25/31$ (d) $31/42$

- 52.** Two numbers are in the ratio 2 : 3. If 9 is added to each number, they will be in the ratio 3 : 4. What is the product of the two numbers? ☑ 2012 I

(a) 360 (b) 480 (c) 486 (d) 512

- 53.** A milkman bought 15 kg of milk and mixed 3 kg of water in it. If the price per kg of the mixture becomes ₹ 22, what is cost price of the milk per kg? ☑ 2012 II

(a) ₹ 28.00 (b) ₹ 26.40 (c) ₹ 24.00 (d) ₹ 22.60

- 54.** Sex ratio is defined as the number of females per 1000 males. In a place, the total inhabitants are 1935000, out of which 935000 are females. What is the sex ratio for the place? ☑ 2012 II

(a) 935 (b) 1000
(c) 1935 (d) 9350

- 55.** In a certain school, the ratio of boys to girls is 7 : 5. If there are 2400 students in the school, then how many girls are there? ☑ 2013 I

(a) 500 (b) 700 (c) 800 (d) 1000

- 56.** If $A : B = 2 : 3$, $B : C = 5 : 7$ and $C : D = 3 : 10$, then what is $A : D$ equal to? ☑ 2014 I

(a) 1 : 7 (b) 2 : 7 (c) 1 : 5 (d) 5 : 1

- 57.** $(x + y) : (x - y) = 3 : 5$ and $xy =$ positive imply that

(a) x and y are both positive ☑ 2014 II
(b) x and y are both negative
(c) one of them is positive and one of them is negative
(d) no real solution for x and y exists

- 58.** The ratio of ages of A and B is 2 : 5 and the ratio of ages of B and C is 3 : 4. What is the ratio of ages of A, B and C ? ☑ 2014 II

(a) 6 : 15 : 20 (b) 8 : 5 : 3
(c) 6 : 5 : 4 (d) 2 : 15 : 4

- 59.** The height of a tree varies as the square root of its age (between 5 to 17 yr). When the age of the tree is 9 yr, its height is 4 ft. What will be the height of the tree at the age of 16 yr? ☑ 2014 II

(a) 5 ft 4 inch (b) 5 ft 5 inch
(c) 4 ft 4 inch (d) 4 ft 5 inch

- 60.** 16 L of a mixture contains milk and water in the ratio 5 : 3. If 4 L of milk is added to this mixture, the ratio of milk to water in the new mixture would be ☑ 2015 I

(a) 2 : 1 (b) 7 : 3 (c) 4 : 3 (d) 8 : 3

- 61.** If $a:b=3:5$ and $b:c=7:8$, then $2a:3b:7c$ is equal to **2015 II**
 (a) $42:105:320$ (b) $15:21:40$
 (c) $6:15:40$ (d) $30:21:350$
- 62.** The speeds of three buses are in the ratio $2:3:4$. The time taken by these buses to travel the same distance will be in the ratio. **2015 II**
 (a) $2:3:4$ (b) $4:3:2$
 (c) $4:3:6$ (d) $6:4:3$
- 63.** In a mixture of milk and water of volume 30 L, the ratio of milk and water is $7:3$. The quantity of water to be added to the mixture to make the ratio of milk and water $1:2$ is **2015 II**
 (a) 30 (b) 32
 (c) 33 (d) 35
- 64.** The annual incomes of two persons are in the ratio $9:7$ and their expenses are in the ratio $4:3$. If each of them saves ₹ 2000 per year, what is the difference in their annual incomes? **2016 I**
 (a) ₹ 4000 (b) ₹ 4500 (c) ₹ 5000 (d) ₹ 5500
- 65.** If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, then which of the following is/are correct?
 I. $\frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = \frac{d}{a}$ II. $\frac{a^2 + b^2 + c^2}{b^2 + c^2 + d^2} = \frac{a}{d}$
 Select the correct answer using the code given below. **2016 I**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

ANSWERS

1	b	2	a	3	c	4	b	5	c	6	a	7	d	8	d	9	a	10	b
11	c	12	b	13	d	14	c	15	c	16	b	17	d	18	c	19	b	20	b
21	b	22	b	23	b	24	a	25	c	26	c	27	a	28	d	29	b	30	c
31	d	32	c	33	d	34	a	35	b	36	b	37	c	38	d	39	c	40	a
41	b	42	a	43	b	44	a	45	d	46	b	47	d	48	c	49	a	50	c
51	c	52	c	53	b	54	a	55	d	56	a	57	d	58	a	59	a	60	b
61	c	62	d	63	c	64	a	65	a										

HINTS AND SOLUTIONS

- 1.** (b) The required compounded ratio is
 $\frac{2a}{6b} \times \frac{7a}{49b} \times \frac{3a}{12b} = \frac{a^3}{84b^3} = a^3 : 84b^3$
- 2.** (a) The subduplicate ratio of
 $16x^4 : 625y^6$ is
 $\frac{\sqrt{16x^4}}{\sqrt{625y^6}} = \frac{4x^2}{25y^3} = 4x^2 : 25y^3$
- 3.** (c) Duplicate ratio of $5:8 = \frac{5^2}{8^2}$
 $\Rightarrow \frac{5^2}{8^2} = \frac{x+5}{3x+4} \Rightarrow \frac{25}{64} = \frac{x+5}{3x+4}$
 $\Rightarrow 25(3x+4) = 64(x+5)$
 $\Rightarrow 75x + 100 = 64x + 320$
 $\Rightarrow 75x - 64x = 320 - 100$
 $\Rightarrow 11x = 220$
 $\therefore x = \frac{220}{11} = 20$
- 4.** (b) Given, $x:6::5:3$
 $\therefore \frac{x}{6} = \frac{5}{3} \Rightarrow 3x = 30 \Rightarrow x = \frac{30}{3} = 10$
- 5.** (c) Let the third proportional to 9 and 12 be x .
 $\therefore 9:12::12:x$
 $\Rightarrow \frac{9}{12} = \frac{12}{x} \Rightarrow 9x = 144$
 $\therefore x = \frac{144}{9} = 16$
- 6.** (a) Mean proportional between $(2 + \sqrt{3})$ and $(8 - \sqrt{48})$
 $= \sqrt{(2 + \sqrt{3})(8 - \sqrt{48})}$
 $= \sqrt{(2 + \sqrt{3})4(2 - \sqrt{3})}$
 $= 2\sqrt{(2 + \sqrt{3})(2 - \sqrt{3})}$
 $= 2\sqrt{4 - 3} = 2 \times 1 = 2$
 $[\because (a-b)(a+b) = a^2 - b^2]$
- 7.** (d) Let fourth proportional be x , then
 $7:11::14:x$
 $\Rightarrow \frac{7}{11} = \frac{14}{x} \Rightarrow 7x = 11 \times 14$
 $\Rightarrow x = 2 \times 11$
 $\therefore x = 22$
- 8.** (d) Let the consequent be x .
 $\therefore 35:x = 7:8$
 Then, $\frac{35}{x} = \frac{7}{8}$ [by given condition]
 $\Rightarrow 7x = 35 \times 8 \therefore x = \frac{35 \times 8}{7} = 40$
- 9.** (a) Given, $x:y=1:3$, $y:z=5:k$,
 $z:t=2:5$ and $t:x=3:4$
 $\frac{x}{y} \times \frac{y}{z} \times \frac{z}{t} \times \frac{t}{x} = 1$
 $\Rightarrow \frac{1}{3} \times \frac{5}{k} \times \frac{2}{5} \times \frac{3}{4} = 1 \Rightarrow k = \frac{1}{2}$
- 10.** (b) Given, $\frac{x}{y} = 25 \left(\frac{y}{x} \right)$
 $\Rightarrow \frac{x^2}{y^2} = \frac{25}{1} \Rightarrow \frac{x}{y} = \frac{5}{1}$ or $5:1$
- 11.** (c) According to the question,
 $p\% \text{ of } x = t \text{ (} q\% \text{ of } y)$
 $\Rightarrow \frac{xp}{100} = \frac{yq}{100} \times t$
 $\therefore x:y = qt:p$

12. (b) $\frac{4ay - 3bx}{5ax - 2by}$ [dividing numerator and denominator by by]

$$\begin{aligned} &= \frac{\frac{4ay}{by} - \frac{3bx}{by}}{\frac{5ax}{by} - \frac{2by}{by}} = \frac{4\left(\frac{a}{b}\right) - 3\left(\frac{x}{y}\right)}{5\left(\frac{a}{b}\right)\left(\frac{x}{y}\right) - 2} \\ &= \frac{4\left(\frac{2}{3}\right) - 3\left(\frac{3}{4}\right)}{5\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) - 2} = \frac{\frac{8}{3} - \frac{9}{4}}{\frac{5}{2} - 2} \\ &= \frac{\frac{32 - 27}{12}}{\frac{5 - 4}{2}} = \frac{12}{1} = \frac{5}{12} \times \frac{2}{1} = \frac{5}{6} \end{aligned}$$

13. (d) Given, $\frac{x^3 + 3x}{3x^2 + 1} = \frac{341}{91}$
By componendo and dividendo rule,
 $\frac{x^3 + 3x + 3x^2 + 1}{x^3 + 3x - 3x^2 - 1} = \frac{341 + 91}{341 - 91} = \frac{432}{250}$

$$\Rightarrow \frac{(x+1)^3}{(x-1)^3} = \frac{216}{125}$$

[$\because (a+b)^3 = a^3 + b^3 + 3ab(a+b)$]

On cube roots both sides, we get
 $\Rightarrow \frac{x+1}{x-1} = \frac{6}{5} \Rightarrow 5(x+1) = 6(x-1)$

$$\therefore x = 11$$

14. (c) Let the number be x

$$\begin{aligned} \therefore \frac{49+x}{68+x} &= \frac{3}{4} \\ \Rightarrow 196 + 4x &= 204 + 3x \\ \therefore x &= 8 \end{aligned}$$

15. (c) Let the three parts be A, B and C .
By given condition,

$$\begin{aligned} \frac{1}{2}A &= \frac{1}{3}B = \frac{1}{6}C = x \quad [\text{say}] \\ \therefore \frac{1}{2}A &= x \Rightarrow A = 2x \\ \frac{1}{3}B &= x \Rightarrow B = 3x \\ \frac{1}{6}C &= x \Rightarrow C = 6x \end{aligned}$$

As, $A + B + C = 1870$
[by given condition]

$$\begin{aligned} \therefore 2x + 3x + 6x &= 1870 \\ \Rightarrow 11x &= 1870 \\ \therefore x &= \frac{1870}{11} = 170 \end{aligned}$$

$\therefore$ Third part i.e. $C = 6 \times 170 = 1020$

16. (b) LCM of 20, 25, 30, 15 is 300.

$$\begin{aligned} \text{So, } \frac{7}{20} &= \frac{7}{20} \times \frac{15}{15} = \frac{105}{300} \\ \Rightarrow \frac{13}{25} &= \frac{13}{25} \times \frac{12}{12} = \frac{156}{300} \end{aligned}$$

$$\Rightarrow \frac{17}{30} = \frac{17}{30} \times \frac{10}{10} = \frac{170}{300}$$

$$\text{and } \frac{11}{15} = \frac{11}{15} \times \frac{20}{20} = \frac{220}{300}$$

As, $\frac{105}{300}$ is the smallest, so $\frac{7}{20}$ or, 7 : 20 is the smallest ratio.

17. (d) Let the numbers be A and B .

$$\therefore 0.5A = 0.07B \quad [\text{by given condition}]$$

$$\Rightarrow \frac{A}{B} = \frac{0.07}{0.5} = \frac{7}{50} \text{ i.e. } 7 : 50$$

18. (c) Let the ratio be x and $(x + 40)$.

Then, by given condition, $\frac{x}{x+40} = \frac{2}{7}$

$$\Rightarrow 7x = 2x + 80 \Rightarrow x = 16$$

So, the required ratio is 16 : 56.

19. (b) Given, $P : Q = \frac{3}{5} : \frac{5}{7}$... (i)

and $Q : R = \frac{3}{4} : \frac{2}{5}$... (ii)

From Eq. (i), $P : Q = \frac{3}{5} \times \frac{3}{4} : \frac{5}{7} \times \frac{3}{4}$
 $= \frac{9}{20} : \frac{15}{28}$... (iii)

From Eq. (ii), $Q : R = \frac{3}{4} \times \frac{5}{7} : \frac{2}{5} \times \frac{5}{7}$
 $= \frac{15}{28} : \frac{2}{7}$... (iv)

From Eqs. (iii) and (iv),

$$P : Q : R = \frac{9}{20} : \frac{15}{28} : \frac{2}{7}$$

20. (b) Let two numbers be $3x$ and $5x$, respectively.

According to the question,

$$(3x - 9) : (5x - 9) :: 12 : 23$$

$$\Rightarrow \frac{3x-9}{5x-9} = \frac{12}{23}$$

$$\Rightarrow 69x - 207 = 60x - 108$$

$$69x - 60x = 207 - 108 \Rightarrow 9x = 99$$

$$\therefore x = 11$$

$$\therefore \text{Second number} = 5x = 5 \times 11 = 55$$

21. (b) Let the numbers are a and b and its mean proportional is x

$$\therefore a : x :: x : b$$

$$a : 28 :: 28 : b \quad (\because x = 28)$$

$$\Rightarrow 28^2 = ab \quad \dots (i)$$

Again, $a : b :: b : 224$

$$\Rightarrow b^2 = a \times 224 \Rightarrow a = \frac{b^2}{224} \quad \dots (ii)$$

From Eq. (i) say,

$$28^2 = \frac{b^2}{224} \times b$$

$$\Rightarrow b^3 = 28 \times 28 \times 224$$

$$\Rightarrow b = 56$$

Again from Eq. (i) say,

$$a = \frac{28 \times 28}{56} = 14$$

22. (b) As, $A : (B + C) = 2 : 9$ [given]

$$\text{Sum of ratios} = 2 + 9 = 11$$

$$\text{So, } A's \text{ part} = 770 \times \frac{2}{11} = ₹ 140$$

23. (b) Given, $x \propto y$ and $x \propto \frac{1}{z^2}$

$$\text{Now, } x \propto \frac{y}{z^2} \Rightarrow x = \frac{ky}{z^2}$$

$$\therefore x = 10, y = 4 \text{ and } z = 14$$

$$\therefore 10 = \frac{k \cdot 4}{196} \Rightarrow k = \frac{1960}{4} = 490$$

Now, $z = 7$ and $y = 16$, then

$$x = \frac{490 \times 16}{7 \times 7} = 160$$

24. (a) $\frac{x}{1} = \frac{\sqrt{m+3n} + \sqrt{m-3n}}{\sqrt{m+3n} - \sqrt{m-3n}}$

By componendo and dividendo rule,

$$\begin{aligned} \frac{x+1}{x-1} &= \frac{\sqrt{m+3n} + \sqrt{m-3n}}{\sqrt{m+3n} - \sqrt{m-3n}} \\ &\quad + \frac{\sqrt{m+3n} - \sqrt{m-3n}}{\sqrt{m+3n} + \sqrt{m-3n}} \\ \Rightarrow \frac{x+1}{x-1} &= \frac{\sqrt{m+3n}}{\sqrt{m-3n}} \end{aligned}$$

On squaring both sides, we get

$$\frac{(x+1)^2}{(x-1)^2} = \frac{m+3n}{m-3n}$$

Again applying componendo and dividendo rule,

$$\begin{aligned} \frac{(x+1)^2 + (x-1)^2}{(x+1)^2 - (x-1)^2} &= \frac{m+3n+m-3n}{m+3n-m+3n} \\ \Rightarrow \frac{2(x^2+1)}{2(2x)} &= \frac{2m}{2(3n)} \end{aligned}$$

$$\Rightarrow \frac{x^2+1}{2x} = \frac{m}{3n} \Rightarrow 3n(x^2+1) = 2mx$$

$$\therefore 3nx^2 - 2mx + 3n = 0$$

25. (c) Given, $p + r = 2q \Rightarrow 2 = \frac{p+r}{q}$

$$\text{and } \frac{1}{q} + \frac{1}{s} = \frac{2}{r}$$

$$\Rightarrow \frac{1}{q} + \frac{1}{s} = \frac{p+r}{qr} \Rightarrow \frac{s+q}{sq} = \frac{p+r}{qr}$$

$$\Rightarrow r(s+q) = s(p+r)$$

$$\Rightarrow rq = sp \Rightarrow \frac{p}{q} = \frac{r}{s}$$

$$\therefore p : q = r : s$$

26. (c) As, given $q^2 = pr$
- $$\frac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}} = \frac{p^2 - q^2 + r^2}{\frac{1}{p^2} - \frac{1}{pr} + \frac{1}{r^2}}$$
- $$= \frac{p^2 - q^2 + r^2}{\frac{(p^2 r^2)(p^2 - q^2 + r^2)}{r^2 - pr + p^2}} = \frac{(p^2 r^2)(p^2 - q^2 + r^2)}{(r^2 - pr + p^2) p^2 r^2}$$
- $$= \frac{(p^2 r^2)(p^2 - pr + r^2)}{(r^2 - pr + p^2)}$$
- $$= p^2 r^2 = (pr)^2 = (q^2)^2 = q^4$$
27. (a) Let monthly incomes of A's and B's be $4x$ and $3x$ respectively
Each saving = ₹ 600
∴ Income - saving = Expenditure
∴ $\frac{4x - 600}{3x - 600} = \frac{3}{2}$
⇒ $2(4x - 600) = 3(3x - 600)$
 $8x - 1200 = 9x - 1800$
⇒ $1800 - 1200 = 9x - 8x$ ⇒ $x = 600$
∴ A's income = $4x = 4 \times 600 = ₹ 2400$
28. (d) Let three numbers be $3x$, $2x$ and $5x$, respectively.
According to the question,
 $(3x)^2 + (2x)^2 + (5x)^2 = 1862$
⇒ $9x^2 + 4x^2 + 25x^2 = 1862$
⇒ $x^2 = \frac{1862}{38} = 49$ ∴ $x = 7$
Hence, the numbers are 3×7 , 2×7 and 5×7 i.e. 21, 14 and 35.
29. (b) Speed $\propto \frac{1}{\text{Time}}$
∴ Required ratio = $\frac{1}{4} : \frac{1}{3} : \frac{1}{2} = 3 : 4 : 6$
30. (c) Let the ages of A and B be $2x$ yr and $5x$ yr, respectively.
After 8 yr age of A = $(2x + 8)$ yr
and age of B = $(5x + 8)$ yr
According to the question, $\frac{2x + 8}{5x + 8} = \frac{1}{2}$
⇒ $4x + 16 = 5x + 8$
⇒ $x = 8$
∴ Difference between their present ages
= $5x - 2x = 3x = 3 \times 8 = 24$ yr
31. (d) By given condition,
 $y = lx + \frac{m}{x}$... (i)
where, l and m are proportionality constants.
when $y = 6$ and $x = 4$,
then $6 = 4l + \frac{m}{4}$
⇒ $16l + m = 24$... (ii)

when $y = \frac{10}{3}$ and $x = 3$,
then $\frac{10}{3} = 3l + \frac{m}{3} \Rightarrow 9l + m = 10$... (iii)
On solving Eqs. (ii) and (iii),
 $l = 2$ and $m = -8$
From Eq. (i), $y = 2x - \frac{8}{x}$

32. (c) Given, $A : B : C = \frac{1}{5} : \frac{1}{6} : \frac{1}{10}$
 $= 6 : 5 : 3$
∴ Share of A = $\frac{6}{6 + 5 + 3} \times 8400$
 $= \frac{6}{14} \times 8400 = ₹ 3600$
33. (d) Let the amount be ₹ x .
In first condition, Q's part = $\frac{5x}{5 + 3} = \frac{5}{8}x$
In second condition, Q's part
 $= \frac{3x}{2 + 3} = \frac{3}{5}x$
By given condition, $\frac{5}{8}x - \frac{3}{5}x = 10$
⇒ $\frac{x}{40} = 10$ ⇒ $x = ₹ 400$
34. (a) Remaining mixture = $20 - 6 = 14$ L
Quantity of milk = $14 \times \frac{4}{7} = 8$ L
∴ Water contents = $14 - 8 = 6$ L
∴ Milk in new mixture
= $8 + 6 = 14$ L and water = 6 L
∴ Ratio of milk and water in new mixture = $\frac{14}{6} = \frac{7}{3}$
Hence, the required ratio 7 : 3.
35. (b) Let x be the part of milk in new mixture.
By law of mixture,
∴ $\frac{\frac{3}{8} - x}{x - \frac{1}{4}} = \frac{3}{2} \Rightarrow \frac{3}{4} - 2x = 3x - \frac{3}{4}$
⇒ $5x = \frac{6}{4} = \frac{3}{2}$ ∴ $x = \frac{3}{10}$
Required ratio = $\frac{3}{10 - 3} = \frac{3}{7}$ or, 3 : 7
36. (b) Let quantities of milk and water be $5x$ and x L, respectively.
According to the question, $\frac{5x}{x + 5} = \frac{5}{2}$
⇒ $10x = 5x + 25$
∴ $x = 5$
Hence, the quantity of milk in the original mixture = $5 \times 5 = 25$ L

37. (c) Given, $x \propto y^m$... (i)
 $y \propto z^n$... (ii)
 $x \propto z^p$... (iii)
On putting the values of x and y from Eqs. (ii) and (iii) in Eq. (i), we get
 $z^p \propto (z^n)^m \Rightarrow z^p \propto z^{nm}$
∴ $p = mn$
38. (d) Let the initial and final salary be ₹ $22x$ and ₹ $25x$, respectively.
and let initial and final number of employees be $3y$ and $2y$, respectively.
∴ Present bill = Final salary $\times$ Final number of employees
 $5000 = 25x \times 2y$
⇒ $\frac{5000}{50} = xy \Rightarrow xy = 100$
∴ Original bill = Initial salary $\times$ Initial number of employees
 $= 22x \times 3y$
 $= 66xy = 66 \times 100 = ₹ 6600$
39. (c) Given,
 $A : B = x : 8$ and $B : C = 12 : z$
⇒ $A : C = \frac{A}{C} = \frac{A}{B} \times \frac{B}{C} = \frac{x}{8} \times \frac{12}{z} = \frac{3x}{2z}$
But $A : C = 2 : 1$
⇒ $\frac{3x}{2z} = \frac{2}{1} \Rightarrow x : z = 4 : 3$
40. (a) Let the number of coins of ₹ 1, 50 paise and 10 paise be $3x$, $8x$ and $10x$, respectively
According to the question,
 $\frac{3x}{1} + \frac{8x}{2} + \frac{10x}{10} = 112$
⇒ $3x + 4x + x = 112$
⇒ $x = \frac{112}{8} = 14$
∴ Number of 50 paise coins
= $14 \times 8 = 112$
41. (b) Let the number of boys and girls be x and y , respectively.
∴ Total number of students = $(x + y)$
According to the question,
Number of boys = Number of girls
+ Total Number of students $\times 12\%$
 $x = y + (x + y) \times 12\%$
 $x - y = \frac{(x + y) \times 12}{100}$
⇒ $25x - 25y = 3x + 3y$
⇒ $22x = 28y$
∴ $x : y = 14 : 11$
42. (a) Let C gets ₹ x .
Now, by given conditions,
Amount of B = ₹ $(x + 8)$
Amount of A = ₹ $(x + 15)$